Regression

Population vs sample: what's the difference?

A population is the entire group that you want to draw conclusions about.

A sample is the specific group that you will collect data from. The size of the sample is always less than the total size of the population.

Note: In research, a population doesn't always refer to people. It can mean a group containing elements of anything you want to study, such as objects, events, organizations, countries, species, organisms, etc.

Etymology

"Linear" means line. The word Regression came from a 19th-Century Scientist, Sir Francis Galton, who coined the term "regression toward mediocrity" (in modern language, that's regression to the mean).

Definition

- Regression is a well-known statistical technique to model the predictive relationship between one or more independent variables and one dependent variables.
- Regression analysis is a statistical method that helps us to analyze and understand the relationship between two or more variables of interest.
- The process that is adapted to perform regression analysis helps to understand which factors (variables or features) are important, which factors can be ignored and how they are influencing each other.

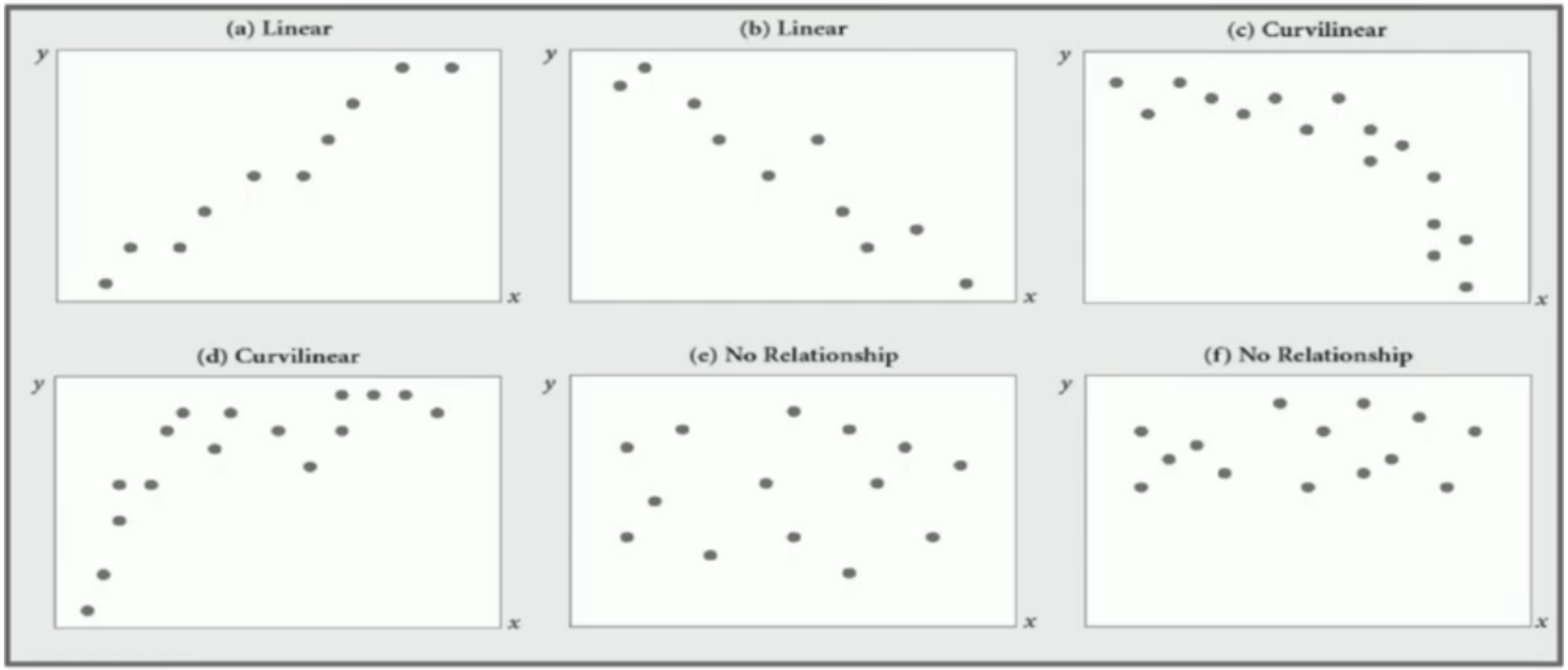
For the regression analysis is be a successful method, we understand the following terms:

- **Dependent Variable:** This is the variable that we are trying to understand or forecast.
- **Independent Variable:** These are factors that influence the analysis or target variable and provide us with information regarding the relationship of the variables with the target variable.

- The objective is to find the best-fitting curve for a dependent variable in a multidimensional space, with each independent variable being a dimension.
- The curve could be a straight line or it could be a nonlinear curve.
- The quality of fit of the curve to the data can be measured by a coefficient of correlation (r). Which is the square root of the amount of variance explained by the curve.

Types of Relationship

Visual Look at Relationships



Correlations and Relationship

- Statistical relationship is about which elements of data hang together, and which ones hang separately.
- It is about categorizing variables that have a relationship with one another, and categorizing variables that are distinct and unrelated to other variables.
- It is about describing significant positive relationships and significant negative differences.
- Given two numeric variables x and y, the coefficient of correlation r is mathematically computed by the following equation. is the mean of x, and is mean of y.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient

= values of the x-variable in a sample

= mean of the values of the x-variable

= values of the y-variable in a sample

= mean of the values of the y-variable

Given the value of Coefficient of correlation, we can identify the kind of relationship between the two variables.

Correlations

- The first and foremost measure of the strength of a relationship is correlation.
- The strength of a correlation is a quantitative measure that is measured in a normalized range between 0 and 1.
- A correlation of **1 indicates a perfect relationship**, where the two variables are in perfect sync.
- A correlation of **0 indicates that there is no relationship** between the variables.

Relationship

The relationship can be **positive, or it can be inverse relationship**, that is, the variables may move together in the same direction or in the opposite direction.

Therefore, a good measure of correlation is the correlation coefficient, which is the square root of correlation.
This coefficient, called r, can thus range from -1 to +1
An r value of 0 signifies no relationship.

An r value of 1 shows perfect relationship in the same direction, and
An r value of -1 shows a perfect relationship but moving in opposite directions.

----So, the relationship variable is nothing but the square root of coefficient of correlation.

Steps in Regression Analysis

The key steps for regression are simple:

1. List all the variables available (dependent and independent) for making the model.
2. Establish a dependent variable of interest.
3. Examine visual (if possible) relationship between variable of interest.
4. Find a way to predict dependent variable using the other variables.

In the second step we need to differentiate between the dependent and independent variables.

In the third step we need to establish the visual relationship between the dependent and independent variables.

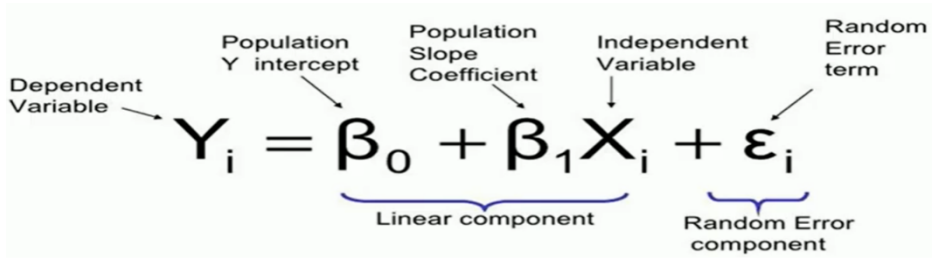
If you are able to establish the visual relationship, then no need to for the fourth step and this will be end of this particular regression modeling.

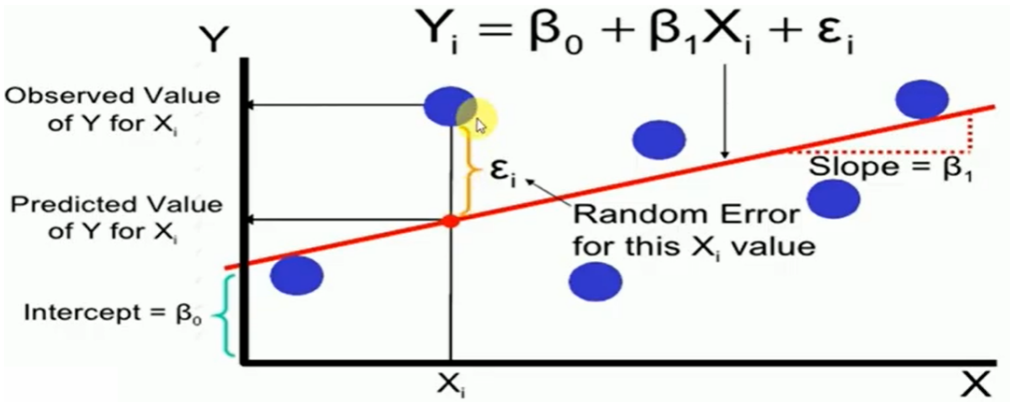
But if you are unable to establish the visual relationship then we can go for the 4th step. In this step we need to identify some way where we can predict the value of dependent variable given the independent variable.

The regression model is described as a linear equation:

If you recall from elementary algebra, the equation for a line is y = mx + b.

- is the predicted value of the dependent variable (**y**) for any given value of the independent variable (**x**).
- is the **intercept**, the predicted value of **y** when the **x** is 0.
- is the regression coefficient – how much we expect **y** to change as **x** increases.
- **x** is the independent variable (the variable we expect is influencing **y**).
- **ε** is the **error** of the estimate, or how much variation there is in our estimate of the regression coefficient.





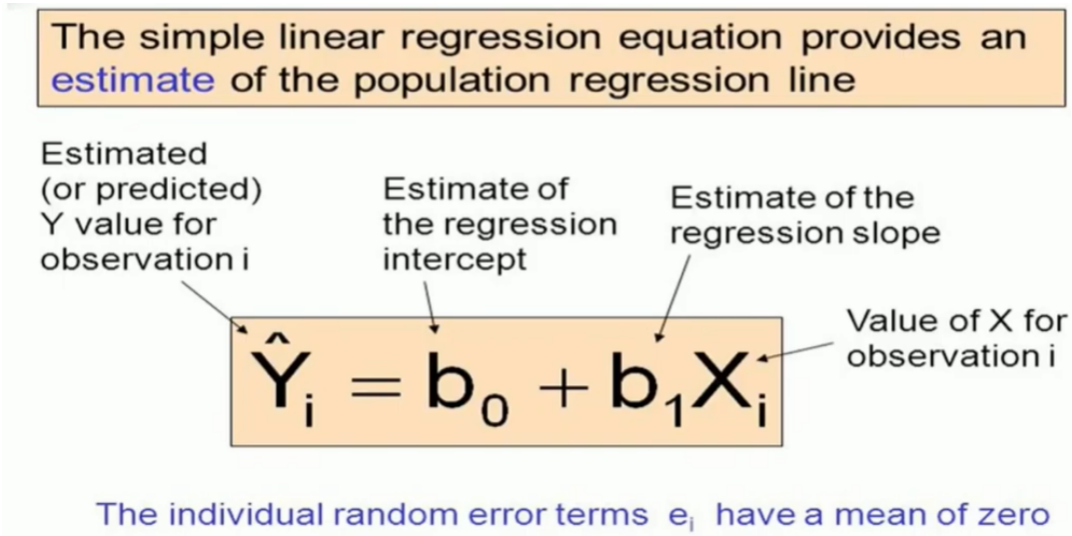
What is the "Best Fitting Line"?

Linear Regression

Regression analysis is used to find equations that fit data. Once we have the regression equation, we can use the model to make predictions. One type of regression analysis **is linear analysis**.

When a **correlation coefficient** shows that data is likely to be able to predict future outcomes and a scatter plot of the data appears to form a straight line, you can use simple linear regression to find a predictive function.

If is the slope of the line and is the y-intercept.



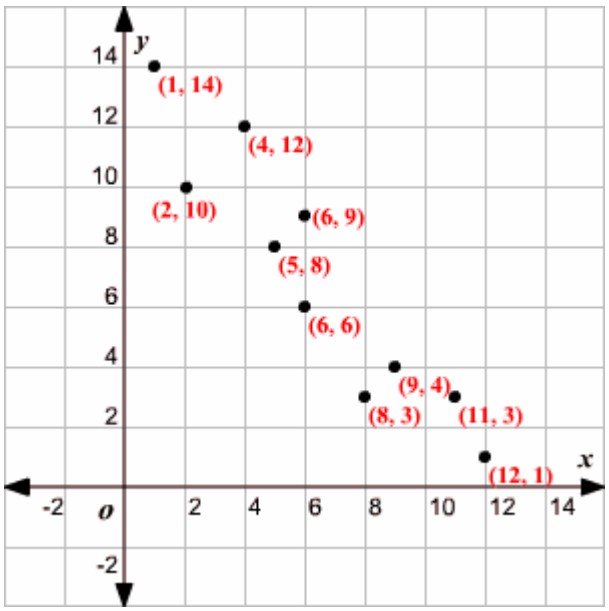
Example

Finding the line of best fit through the least square method:

You can find the equation for the line of best fit using the least square method in four steps.

(8,3),(2,10),(11,3),(6,6),(5,8),(4,12),(12,1),(9,4),(6,9),(1,14)

It may help to see these ordered pairs plotted:



` $\bar{x}$ ` is the symbol used to specify the mean of the x values, and you can calculate it by adding all 10 x values above and then dividing by 10. In the example above, ` $\bar{x} = 6.4$ `.

` $\bar{y}$ ` is the symbol used to specify the mean of the y values. ` $\bar{y} = 7$ ` in the example above.

Next, we need to use both means to find numbers we'll eventually plug into formulas. Specifically, we need ` $x - \bar{x}$ ` and ` $y - \bar{y}$ ` for every value of x and y above. We are going to build a table to make the data easier to read:

x-coordinate	$x - \bar{x}$	y-coordinate	$y - \bar{y}$
8	1.6	3	-4
2	-4.4	10	3
11	4.6	3	-4
6	-0.4	6	-1
5	-1.4	8	1
4	-2.4	12	5
9	2.6	4	-3
6	-0.4	9	2
1	-5.4	14	7
12	5.6	1	-6

We're finally done with the first step! Now, we need to calculate the slope of the line of best fit using the following formula:

$$m = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$m = -\frac{131}{118.4} \approx 1.1$$

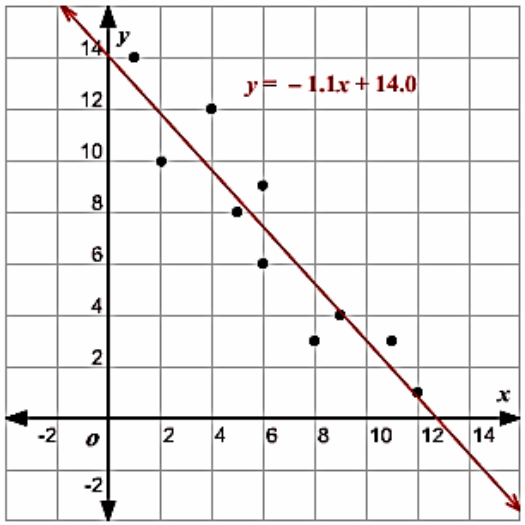
We have our slope! Now, we can plug the slope into the following formula for the y-intercept:

$$b = \bar{y} - m \times \bar{x}$$

$$b = 7 - (-1.1 \times 6.4)$$

$$b \approx 14$$

We now have a slope of -1.1 and a y-intercept of 14, enabling us to write the equation $y = -1.1x + 14$. Note that we rounded both numbers to make them easier to work with. The final step is to draw the line on our scatter plot:



It's a lot of work for a single problem, but you can do it so long as you keep your wits about you and take it one step at a time.

Line of best fit practice question

a. Write an equation for the line of best fit for the following ordered pairs:

$(2,4),(3,5),(5,7),(7,10),(9,15)$

To find the line of best fit, we can use the least squares regression method. However, I'll show you a simplified version of the method to obtain an approximate line.

First, find the means of x and y values:

$$\text{Mean of x} = \frac{2+3+5+7+9}{5} = \frac{26}{5} = 5.2$$

$$\text{Mean of x} = \frac{4+5+7+10+15}{5} = \frac{41}{5} = 8.2$$

Then, find the slope (m):

$$m \approx \frac{\sum(x-\bar{x})(y-y_{\text{mean}})}{\sum(x-\bar{x})^2}$$

For each point (x, y), calculate $(x-\bar{x})(y-y_{\text{mean}})$ and $(x-\bar{x})^2$, then sum the results.

$$m \approx \frac{(2-\bar{x})(4-y_{\text{mean}})+(3-\bar{x})(5-y_{\text{mean}})+\ldots}{(2-\bar{x})^2+(3-\bar{x})^2+\ldots}$$

$$m \approx \frac{-3.2\times(-4.2)+(-2.2)\times(-3.2)+..}{3.2^2+2.2^2+..}$$

$$m \approx \frac{38.4}{25.6} \approx 1.5$$

Now find the y-intercept (b):

$$b = \bar{y} - m \times \bar{x}$$

$$b = 8.2 - 1.5 \times 5.2$$

$$b = 0.3$$

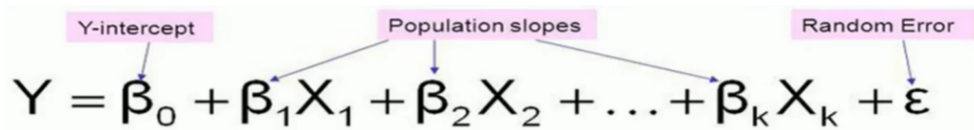
The approximate line of best fit is:

$$y = 1.5x + 0.3$$

MULTIPLE LINEAR REGRESSION

Multiple linear regression analysis is essentially similar to the simple linear model, with the exception that multiple independent variables are used in the model. The mathematical representation of multiple linear regression is:

We need a regression model



$$= b_0 + b_1 x_1 + b_2 x_2$$

Multiple linear regression follows the same conditions as the simple linear model. However, since there are several independent variables in multiple linear analysis, there is another mandatory condition for the model:

- **Non-collinearity:** Independent variables should show a minimum correlation with each other. If the independent variables are highly correlated with each other, it will be difficult to assess the true relationships between the dependent and independent variables.